

Samuel Schapiro | Curriculum Vitae



Research Interests

Artificial intelligence; the science of intelligence; creativity; automated scientific discovery

Education

University of California, Berkeley

Visiting Graduate Student, Theoretical Computer Science

Aug 2024 – Dec 2024

Berkeley, CA

University of Illinois, Urbana-Champaign

Bachelor of Science, Computer Science

Aug 2021 – May 2025

Champaign, IL

· Founded Quant, UIUC's first student-run quant trading fund.

Honors and Awards

Lambda AI Research Grant \$5,000

2026

Best Short Paper Award Association for Computational Creativity

ICCC 2025

Edmund J. James Honor Scholar University of Illinois, Urbana-Champaign

2023

Cozad New Venture Challenge \$35,000

2022

Research Directions

Foundations of Creativity | *Goal:* To uncover fundamental principles of creativity governing minds and machines. Achieved through a combination of conceptual, theoretical work, and empirical study of LLMs.

- **[NeurIPS 2026, ICML Workshop on Creativity 2026]** [2] Studies whether semantic distance tests predict creativity in LLMs. [Key finding: Existing tests do not predict scientific ideation ability.](#)
- **[ICCC 2025, Best Short Paper Award]** [4] Derives a logic for transformational creativity by synthesizing cognitive science and philosophy of science; uses it to explain scientific paradigm shifts.
- **[Preprint]** [3] Proposes a theoretical framework for combinatorial creativity in LLMs that exposes a fundamental tradeoff between novelty and feasibility.

Creative AI Systems | *Goal:* To develop methods that make AI models more creative, and to apply these techniques to scientific discovery.

- **[COLM 2026, ICML Workshop on Creativity 2026]** [1] Introduces CreativityNeuro, a weight steering method that improves LLM divergent thinking by up to 14 human percentile points. [Key finding: Steering LLM creativity in weights, not activations, generalizes further to unseen tasks.](#)
- **[ICCC 2025]** [5] Introduces Spark, a retrieval-augmented LLM idea generator paired with Judge, a model trained on 600K OpenReview reviews, enabling end-to-end scientific idea generation and evaluation.

Positions Held

Prior Computers Founding Research Scientist	Nov 2025 – Mar 2026 Cambridge, MA
Information and Intelligence Group Research Assistant	Nov 2024 – Nov 2025 Champaign, IL & Stony Brook, NY
Simons Institute for the Theory of Computing, UC Berkeley Visiting Graduate Researcher	Aug 2024 – Dec 2024 Berkeley, CA
Trustworthy Machine Learning Group [6] Research Assistant	May 2023 – Dec 2024 Champaign, IL
HAP Capital Quantitative Research Intern	Jun 2022 – Aug 2022 New York, NY

Publications

- [1] **[COLM 2026] S. Schapiro**, Core Francisco Park, Felix Sosa, and Lav R. Varshney. *CreativityNeuro: Steering Language Model Weights to Improve Divergent Thinking*. Under review, Third Conference on Language Modeling (COLM). 2026. url: <https://schapiro.ai/files/CN.pdf>.
- [2] **[NeurIPS 2026] S. Schapiro**, Alexi Gladstone, Jonah Black, and Heng Ji. *Do Semantic Distance Tests Actually Predict Creativity in Large Language Models?* Under review, 40th Conference on Neural Information Processing Systems (NeurIPS). 2026. url: https://schapiro.ai/files/SD_NIPS.pdf.
- [4] **[ICCC 2025, Best Short Paper Award] S. Schapiro**, Jonah Black, and Lav R. Varshney. *Transformational Creativity in Science: A Graphical Theory*. *Proceedings of the 16th International Conference on Computational Creativity (ICCC)*. 2025. arXiv: 2504.18687.
- [3] **[Preprint] S. Schapiro**, Sumuk Shashidhar, Jonah Black, Dilek Hakkani-Tur, and Lav R. Varshney. *Combinatorial Creativity: A New Frontier in Generalization Abilities*. Preprint. 2025. arXiv: 2509.21043.
- [5] **[ICCC 2025]** Aishik Sanyal, **S. Schapiro**, Sumuk Shashidhar, Royce Moon, Lav R. Varshney, and Dilek Hakkani-Tur. *Spark: A System for Scientifically Creative Idea Generation*. *Proceedings of the 16th International Conference on Computational Creativity (ICCC)*. 2025. arXiv: 2504.20090.
- [6] **[Undergraduate Thesis] S. Schapiro** and Han Zhao. *Towards Understanding the Role of Sharpness-Aware Minimization Algorithms for Out-of-Distribution Generalization*. Undergraduate thesis. University of Illinois, Urbana-Champaign, 2024. arXiv: 2412.05169.

Service

Conference Reviewer

- ICML 2026 (Workshop on Creativity & Generative AI)
- ICML 2026
- COLM 2026
- NeurIPS 2026
- NeurIPS 2024 (Workshop on Creativity & Generative AI)

Organizer & Presenter

- Generalization in Philosophy and Psychology Reading Group, 2024

- Optimal Transport Reading Group 2024
- Causal Inference Reading Group 2024
- High Dimensional Probability Reading Group 2023
- Trustworthy Machine Learning Reading Group 2023

Selected Coursework

Machine Learning, Statistical Learning Theory*, Statistical Reinforcement Learning*, Optimization for Learning and Vision*, Applied Machine Learning, Deep Learning, Reinforcement Learning*, Machine Learning Seminar*, Probability & Statistics, Linear Algebra, Real Analysis, Algorithms and Models of Computation, Introduction to Optimization, Discrete Mathematics, Calculus Sequence.

(* graduate-level)